

LESSON 18 / 第十八課

Trees on Engineered Slopes: From Catchment Duty to the Root-Nail Contact

工程邊坡上的樹木：從集水區職責到根-土釘接觸

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

A tree on a registered or engineered slope lives in two residual-risk sentences at once: the slope’s duty and the tree’s duty. Geotechnical publications treat vegetation as cover, surcharge, seepage path or reinforcement — not as an amenity specimen. Soil nails are wounds in the cover. Roots can stitch a facing or concentrate seepage. Removal changes both sentences. Hydraulics on an engineered surface are not hydraulics on a forest floor.

登錄或工程邊坡上的樹同時活在兩句剩餘風險裡：邊坡的職責與樹的職責。岩土出版物把植被當成覆蓋、附加荷載、滲徑或加筋——不是園景標本。土釘是覆蓋中的傷口。根可以縫合護面，也可以集中滲流。移除改變兩句。工程表面上的水力，不是森林地表上的水力。

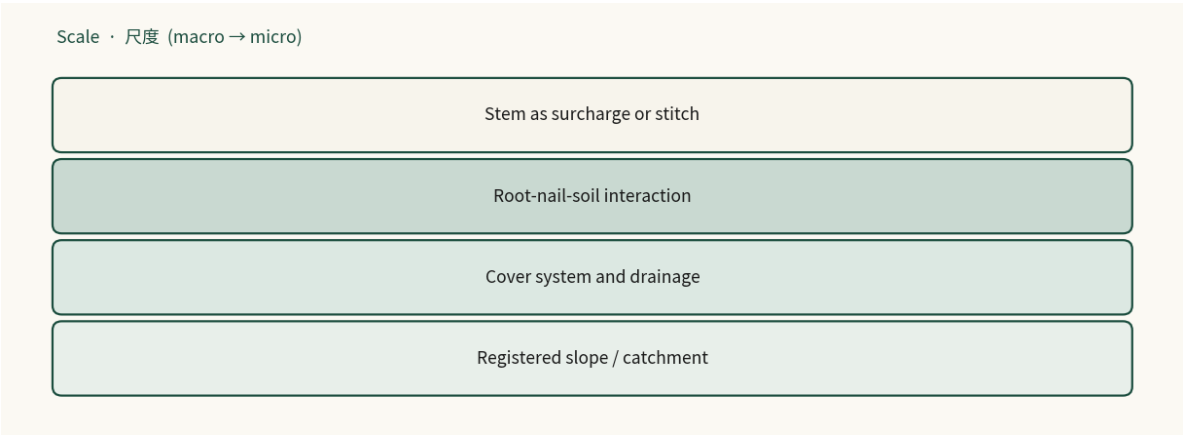


Figure 1. Nested scales from catchment duty to root-nail contact.

圖 1. 由集水區職責到根-土釘接觸的嵌套尺度。



Figure 2. Two duties that do not substitute.

圖2. 不可替代的兩種職責。

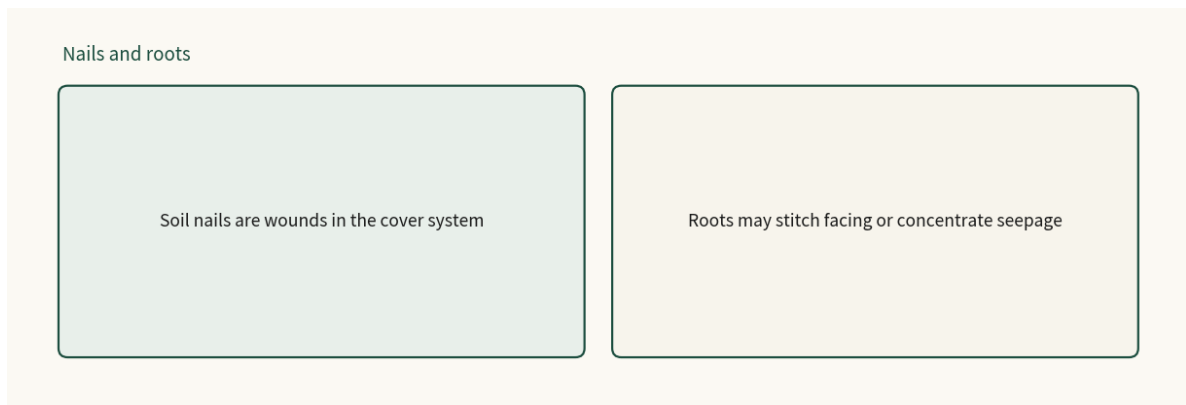


Figure 3. Nails wound the cover; roots stitch or concentrate seepage.

圖3. 土釘傷害覆蓋；根縫合或集中滲流。

1. Scope

1. 範圍

The object is published interaction of trees with engineered slopes. A local slope catalogue is an example of the geotechnical sentence, not the only one.

對象是樹木與工程邊坡的已發表相互作用。地地邊坡目錄是岩土句子的例子，不是唯一例子。

2. Dual sentence

2. 雙重句子

Geotechnical duty asks whether the cover, drainage and surcharge still meet the slope's performance criteria [1,2]. Tree-risk duty asks the target-likelihood-consequence question of a specified part [3]. Vegetation can be cover, reinforcement, surcharge or a seepage path depending on species, density and facing [2]. Those roles do not cancel the tree-risk sentence.

岩土職責問覆蓋、排水與附加荷載是否仍滿足邊坡性能準則 [1,2]。樹木風險職責對指定部位問目標-可能性-後果 [3]。植被按樹種、密度與護面，可以是覆蓋、加筋、附加荷載或滲徑 [2]。那些角色不取消樹木風險句子。

3. Nails, roots, hydraulics

3. 土釘、根、水力

Soil nails and drains are discontinuities in the cover. Roots that follow those discontinuities can stitch or jack. Engineered facings change infiltration and runoff relative to a forest floor; irrigation and storm water must be specified against that surface. Removal rewrites both residual-risk sentences.

土釘與排水管是覆蓋中的不連續。跟隨那些不連續的根可以縫合或頂推。相對森林地表，工程護面改變入滲與徑流；灌溉與雨水必須對該表面規格化。移除改寫兩句剩餘風險。

4. Limits

4. 限度

Root-reinforcement numbers from temperate forest slopes transfer poorly onto short, nailed urban cuttings with thin facing soils.

溫帶森林邊坡的根系加筋數字，難以轉到護面土薄的短土釘城市開挖上。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. A tree on an engineered slope carries:

1. 工程邊坡上的樹承載：

A. One sentence only.

A. 只有一句。

B. Two residual-risk sentences.

B. 兩句剩餘風險。

C. Only P50.

C. 只有 P50。

D. Only a register.

D. 只有名錄。

2. Geotechnical duty concerns:

2. 岩土職責關切：

A. Flower scent.

A. 花香。

B. Cover, seepage and surcharge against slope criteria.

B. 對邊坡準則的覆蓋、滲流與附加荷載。

C. i-Tree only.

C. 只有 i-Tree。

D. Paint.

D. 漆。

3. Soil nails are:

3. 土釘是：

A. Invisible to the cover.

A. 對覆蓋不可見。

B. Wounds / discontinuities in the cover.

B. 覆蓋中的傷口／不連續。

C. Wall 4.

C. 第 4 壁。

D. 10-20-30.

D. 10-20-30。

4. Roots on a facing can:

4. 護面上的根可以：

A. Only decorate.

A. 只裝飾。

B. Stitch or concentrate seepage.

B. 縫合或集中滲流。

C. Raise flash density.

C. 提高閃擊密度。

D. Set SOS.

D. 設定 SOS。

5. Removal:

5. 移除：

A. Leaves the slope sentence unchanged.

A. 邊坡句子不變。

B. Rewrites both sentences.

B. 改寫兩句。

C. Equals t/R.

C. 等於 t/R。

D. Is a mulch rule.

D. 是覆蓋規則。

6. Hydraulics on an engineered surface:

6. 工程表面上的水力：

A. Equal forest-floor hydraulics.

A. 等於森林地表水力。

B. Are not forest-floor hydraulics.

B. 不是森林地表水力。

C. Equal P88.

C. 等於 P88。

D. Equal QTRA.

D. 等於 QTRA。

7. Vegetation roles on slopes:

7. 邊坡上植被的角色：

A. Are always only amenity.

A. 永遠只是園景。

B. Include cover, reinforcement, surcharge, seepage path.

B. 包括覆蓋、加筋、附加荷載、滲徑。

C. Are only carbon.

C. 只有碳。

D. Are only 3-30-300.

D. 只有 3-30-300。

8. Tree-risk method on a slope tree:

8. 邊坡樹上的樹木風險方法：

A. Is abolished.

A. 被取消。

B. Still applies to specified parts and targets.

B. 仍適用於指定部位與目標。

C. Is a soil test.

C. 是土壤檢測。

D. Is a cable.

D. 是鋼纜。

9. Transfer of forest root-reinforcement numbers onto short nailed cuttings:

9. 把森林根系加筋數字轉到短土釘開挖：

A. Is automatic.

A. 自動的。

B. Transfers poorly.

B. 難以轉移。

C. Is Wall 1.

C. 是第 1 壁。

D. Is a flare datum.

D. 是幹基基準。

10. GEO landscape-treatment language is:

10. 岩土景觀處理語言是：

A. The only sentence.

A. 唯一句子。

B. One published cover language; tree-risk is the other.

B. 已發表覆蓋語言之一；樹木風險是另一句。

C. A tomogram.

C. 斷層圖。

D. An inoculant.

D. 接種劑。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Catalogue / inventory systems for registered slopes (comparative).

[2] GEO Publication 1/2011 and related landscape-treatment guidance.

[3] ANSI A300 Part 9 applied on engineered ground.

[4] Stokes A. et al. eco-engineering and root reinforcement.

[5] Gray D.H., Sotir R.B. biotechnical slope protection.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。